Lab 2: Processing steps and Radar Geometry

GRD is not raw SAR

- Sentinel-1 product levels: SLC vs GRD
  - SLC - Single-Look Complex
    - Full resolution, phase included
  - GRD - Ground Range Detected
    - Multi-looked (reduced resolution), projected to ground range (flat map projection)
    - No phase information
- GRD still needs calibration / geocoding / terrain correction depending on use
  - Calibration - converts backscatter intensity to a normalized radar cross-section
    - Essential for quantitative use of SAR data
  - geocoding/terrain correction - geocode the image from slant or ground range into CRS (map)
    - Uses a DEM (digital elevation model) to correct radar geometry distortions

What is in a Sentinel-1 GRD product?

- Amplitude_VV, Amplitude_VH, Intensity_VV, Intensity_VH
  - Amplitude - raw strength of received signal. Useful for visualization
  - Intensity - Amplitude^2 - used for radiometric calibration
- metadata: orbit, pass direction, acquisition time, polarization, acquisition mode (e.g., IW)
- We will want to convert these values to a normalized backscatter value, and georeference geometry
  - Sigma0_V{V,H} = calibrated backscatter quantity

Minimal GRD preprocessing chain

Open GRD

- Apply Orbit File
- Radiometric Calibration
- (optional) Multilooking
- Terrain Correction
- Export GeoTIFF
- We will Analyze in Python

Single Look Complex

Range Looks: 1
Azimuth Looks: 1

Range Spacing: 7.8 m
Azimuth Spacing: 4.0 m

Multilooked

Range Looks: 4
Azimuth Looks: 20

Range Spacing: 31.2 m
Azimuth Spacing: 80.0 m

Multi-looking can reduce noise and give approximately square pixels size, at the expense of spatial resolution

https://step.esa.int/docs/tutorials/S1TBX%20SAR%20Basics%20Tutorial.pdf

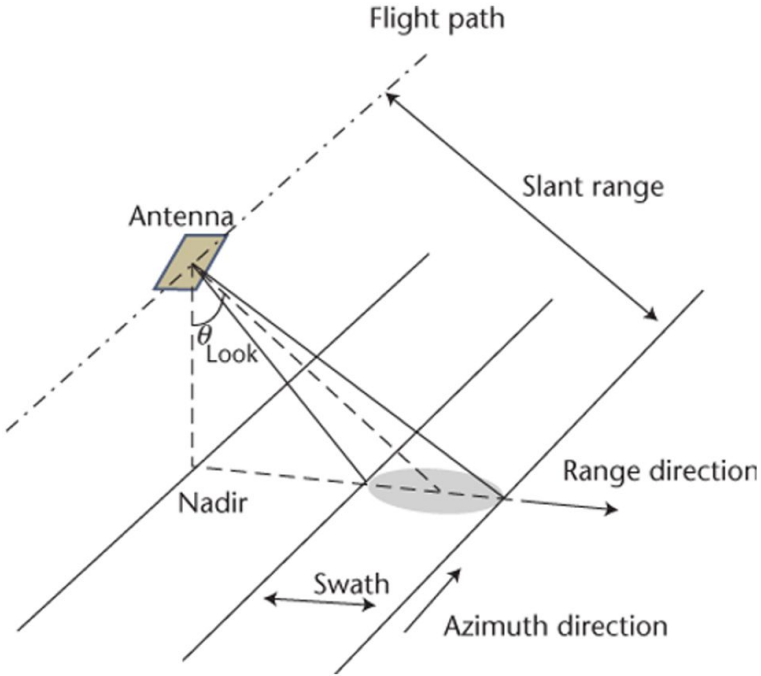
Radar geometry: SAR is side-looking

- SAR is side-looking, not nadir (straight down) looking.
- Range, azimuth, slant range, look angle/incidence angle.
- This is the single most important geometric idea.

Slant Range Distance
- The direct distance between the sensor and the target

Range Direction (Across Track)
- The direction perpendicular to the motion of the radar sensor

Azimuth Direction (Along Track)
- The direction parallel to the motion of the radar platform as it moves over the target area



https://appliedsciences.nasa.gov/sites/default/files/2024-11/SAR_2024_Part1_English_FINAL.pdf

Foreshortening, layover, and shadow

Foreshortening: slope facing radar is compressed

Layover: top of mountain appears before base. With layover, it's not just geometry that gets obscured, it also leads to ambiguity of where the radar returns are coming from (top of mountain vs foreground before mountain).

Shadow: radar beam cannot illuminate slope behind obstacle

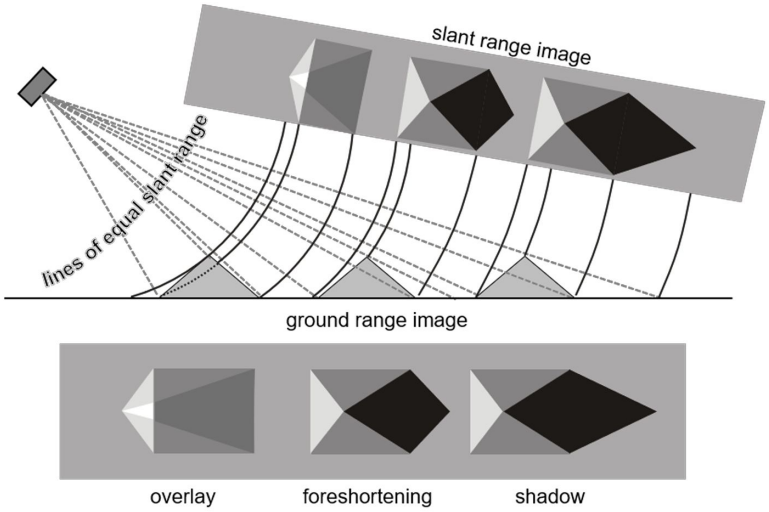


Figure 11: Geometric distortions in radar images (Braun 2019)

Layover and shadow mask output in SNAP

Mask	Meaning	Interpretation
0	Normal	Geometry likely usable
1	Layover	Radar-facing steep terrain. Range ordering distorted
2	Shadow	Not illuminated. No reliable return.
3	Layover + Shadow	Severe terrain-geometry ambiguity. Exclude or flag strongly.

Terrain correction: what it does and does not do

- pre-TC = radar geometry, no normal CRS (coordinate reference system)
- post-TC = map geometry, CRS such as EPSG:4326 or UTM
- DEM is used to project pixels onto terrain

Important distinction:

- **Geometric correction:** moves pixels to map coordinates
- **Radiometric terrain correction:** corrects slope-related brightness effects more rigorously

Terrain Correction makes the data map-like, but it is not magically perfect - distortions (e.g., stretched or compressed pixels) due to terrain will still be present

DEM, incidence angle, and layover/shadow mask

These are optional outputs from SNAP during the terrain correction step:

- DEM: terrain used for correction
- local incidence angle: angle between radar beam and actual slope
- layover/shadow mask: geometry-based estimate of unreliable pixels

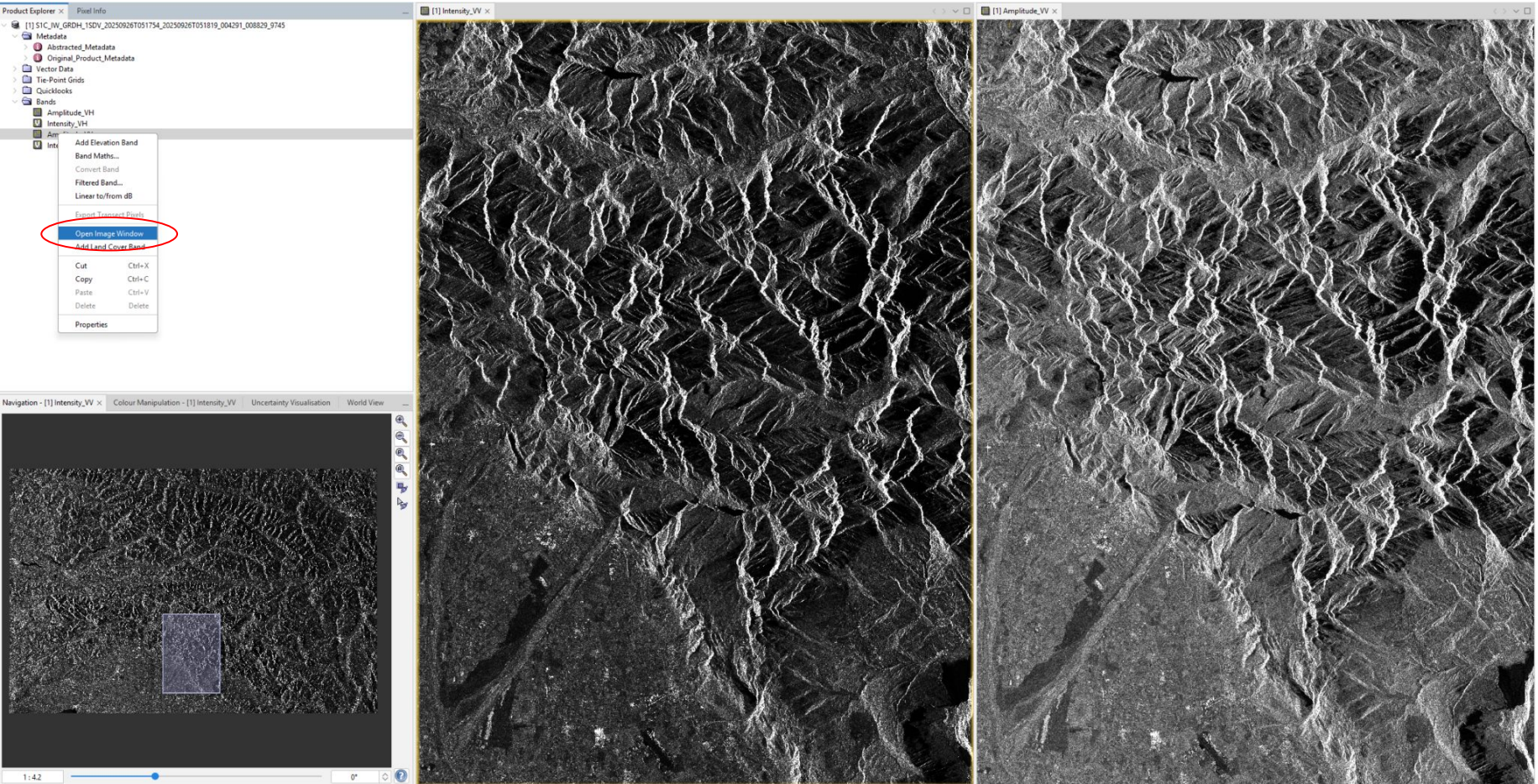
Where to access data:

Copernicus Data Space Ecosystem (CDSE)
https://browser.dataspace.copernicus.eu/

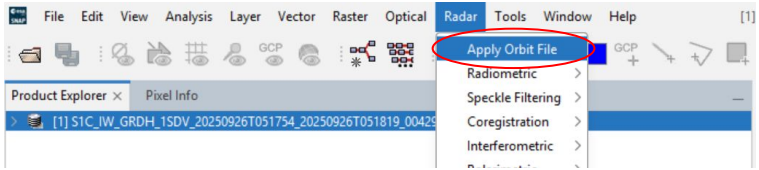
Alaska Satellite Facility - ASF Vertex (some processing tools included, burst level SLC products, zipped bundles for compressed storage)

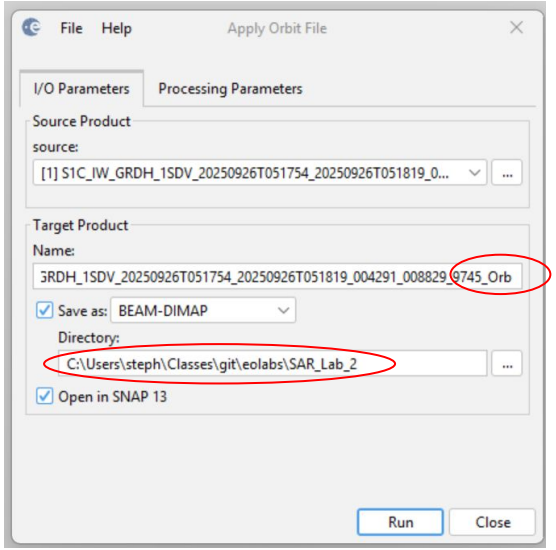
https://search.asf.alaska.edu/

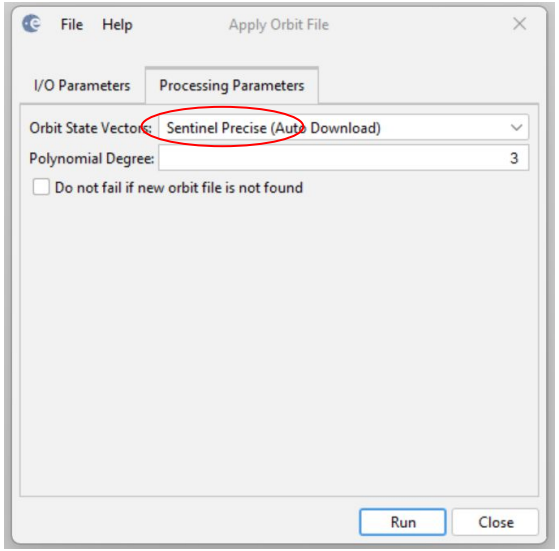
1. Open GRD, explore metadata, Amplitude, Intensity



Apply Orbit File



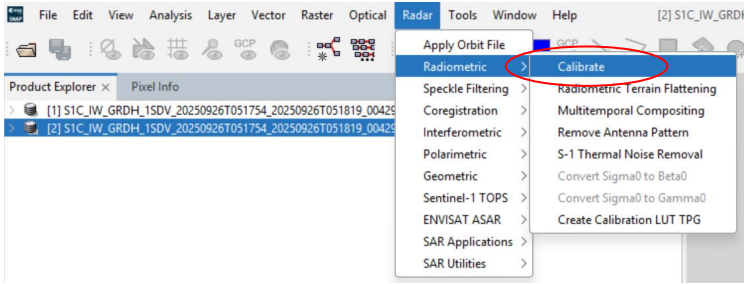


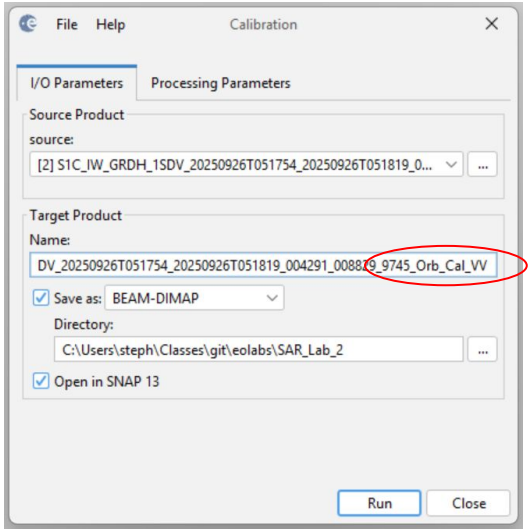


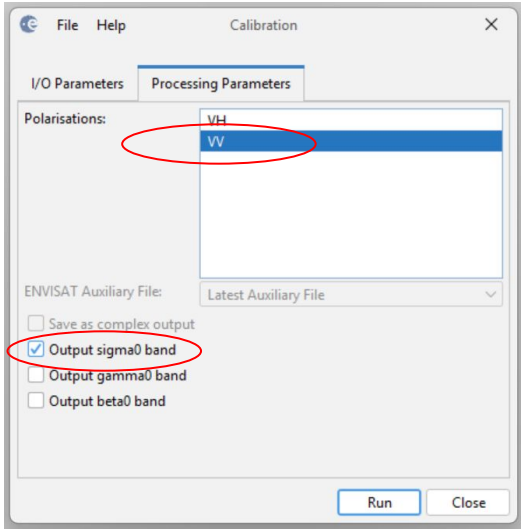
Precise - 20 day latency, 1-5 cm
Restituted - hours latency, ~5-10 cm

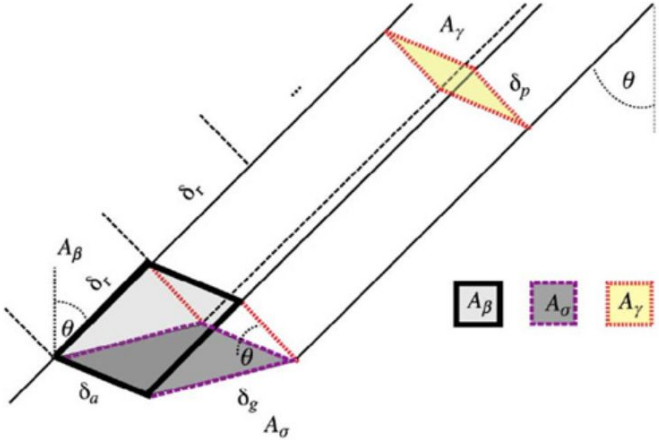
*Always use precise, unless looking at ongoing issues (e.g., disaster response).

Radiometric Calibration





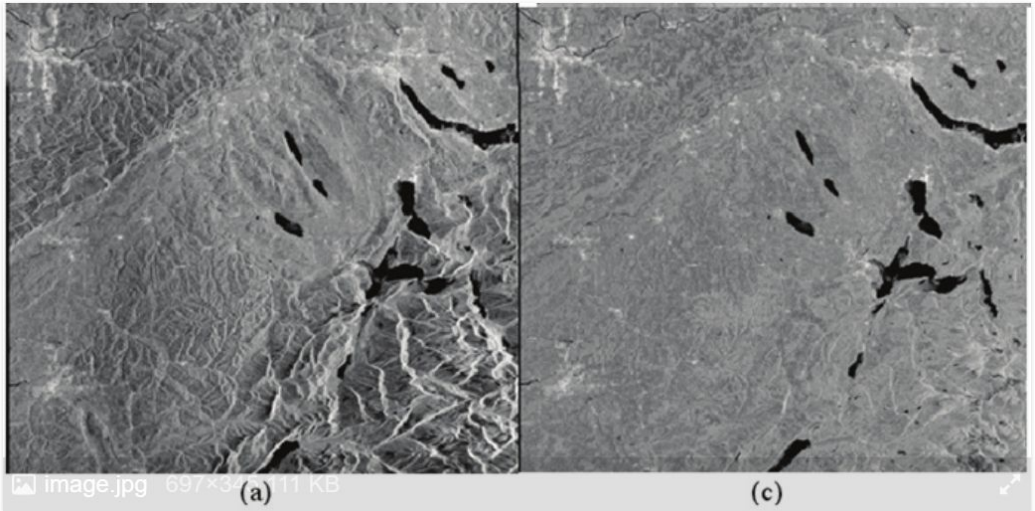




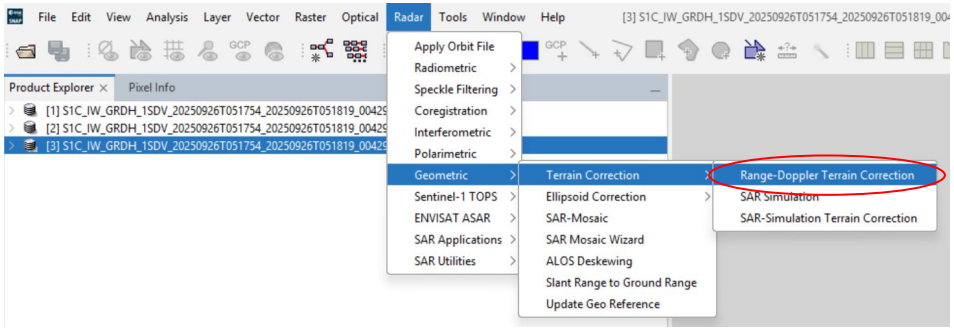
- Sigma0 - normalized radar cross-section, referenced to horizontal ground area
- Gamma0 - backscatter perpendicular to look direction
- Beta0 - normalized to slant range plane, mainly used for further topographic corrections

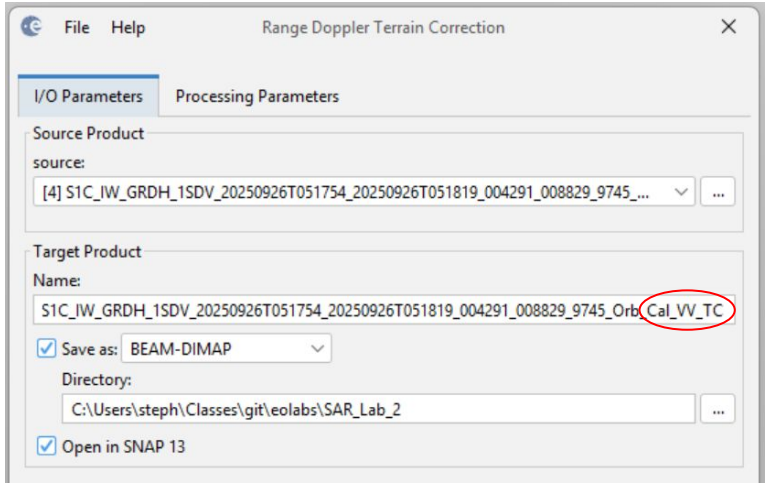
*This lab is learning about terrain and geometry effects, so we'll use sigma0.

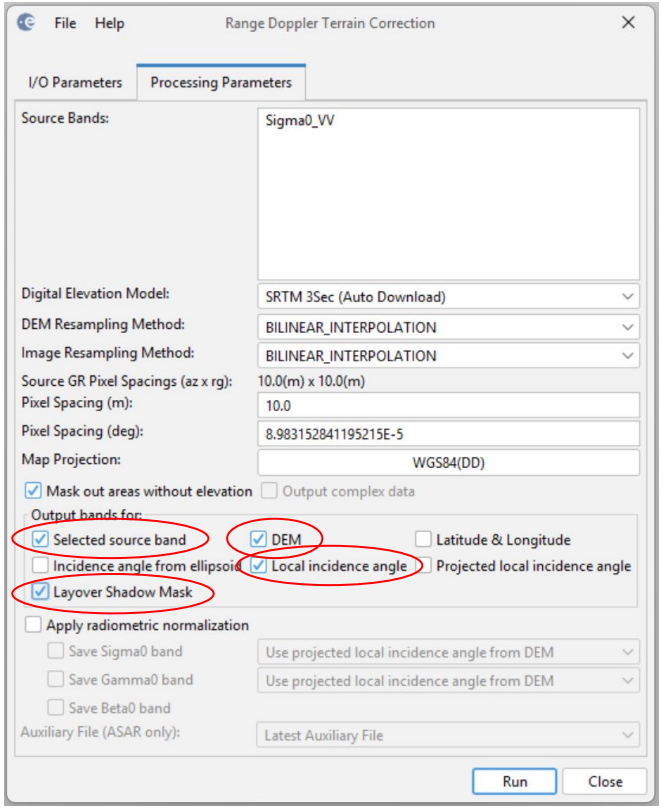
Just an example of full radiometric terrain flattening



Terrain Correction





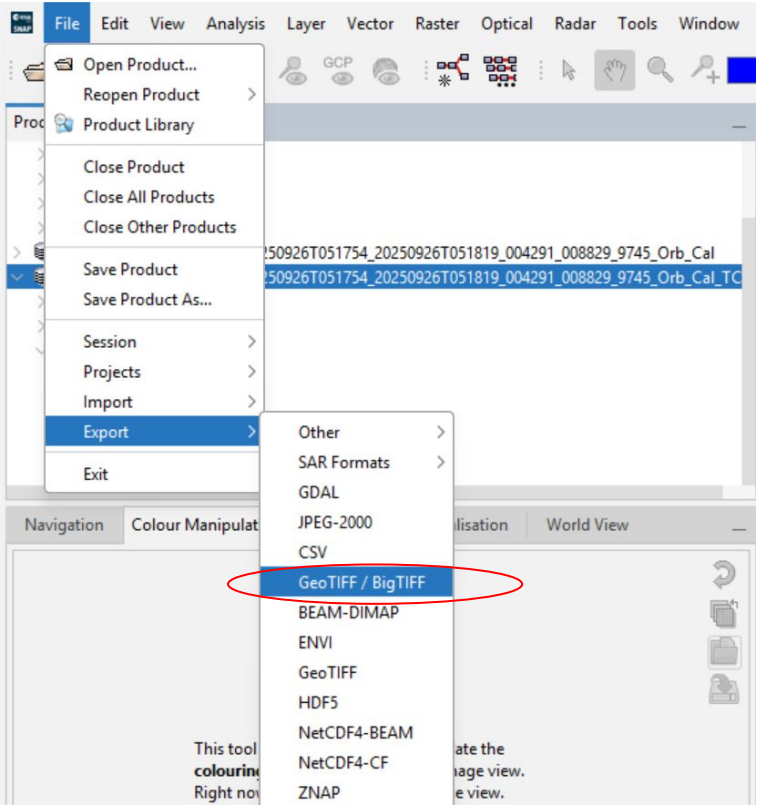


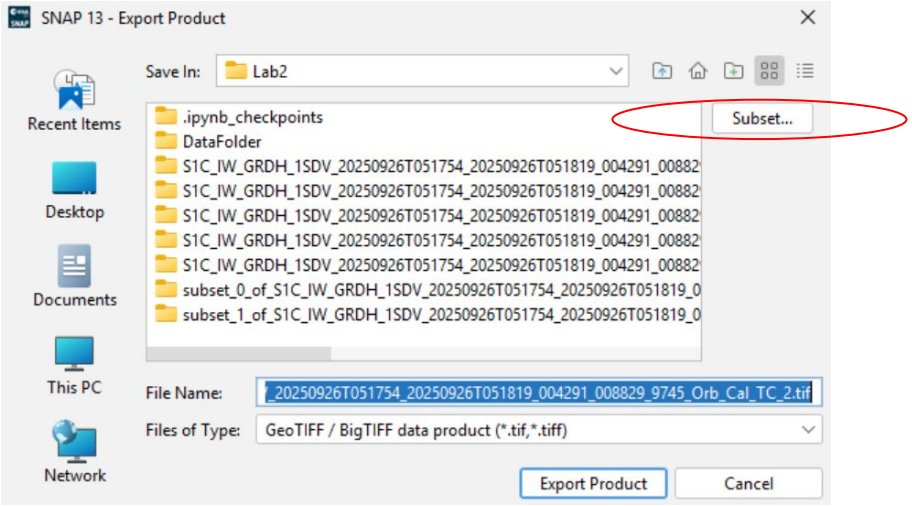
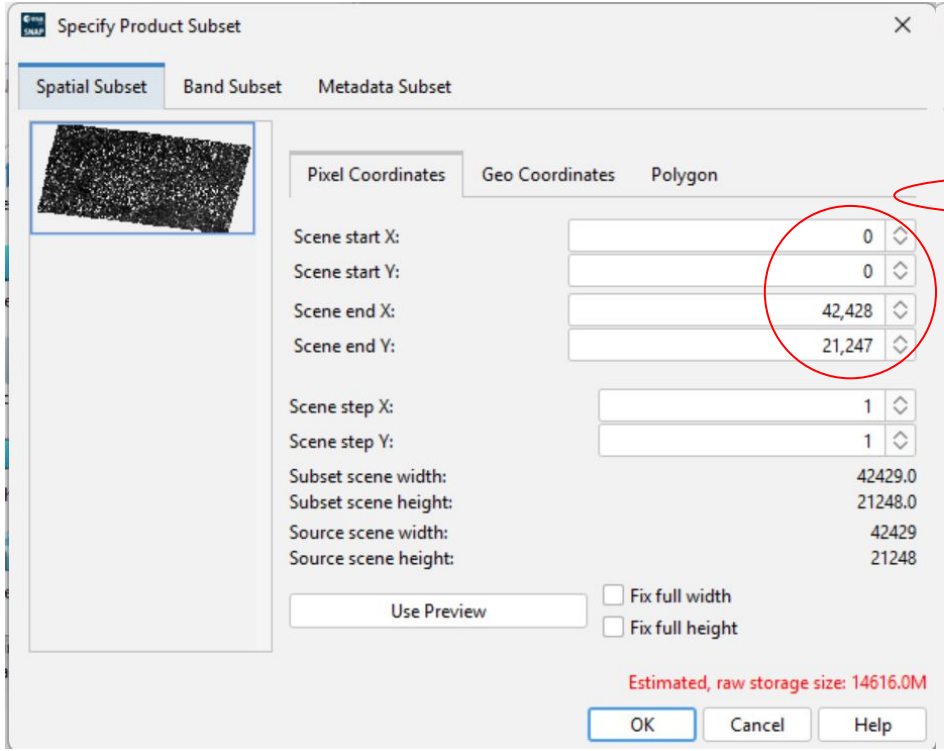
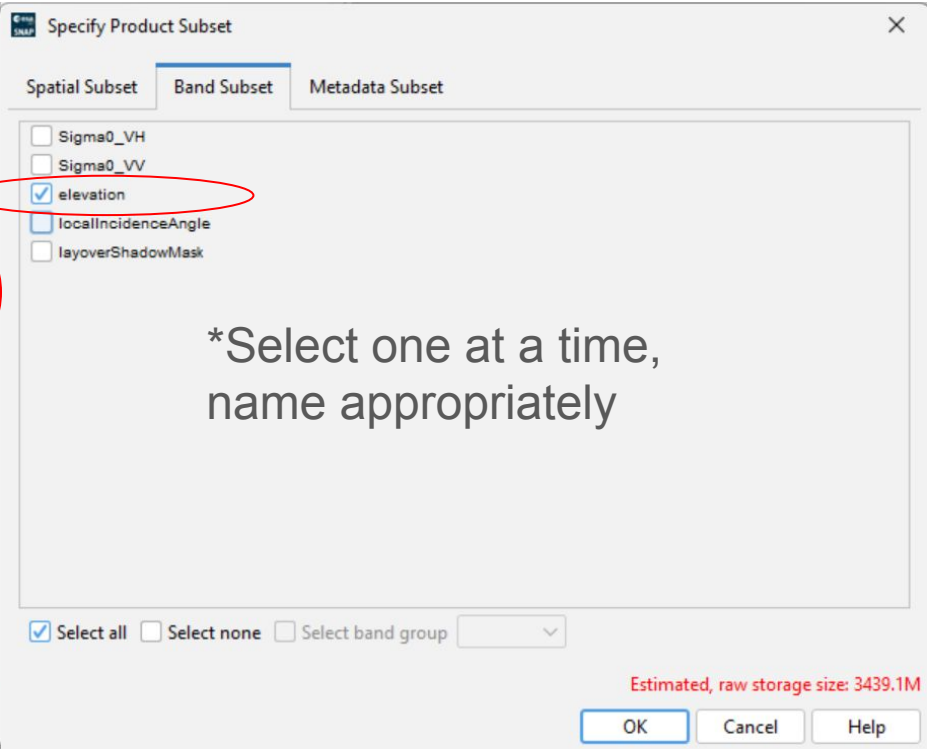
Export as GeoTIFF for analysis in Python

- Pre-TC VV
- Post-TC VV
- DEM
- Local Incidence Angle
- Layover/Shadow mask

*For all the post-TC exports, export them one channel at a time.

Export as GeoTIFF

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